Cisco PE MAR-PE-13 Hardware/Environmental Fault Resolution SOP (ALARM_2961)

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_2961

Description

This Standard Operating Procedure details the steps required to troubleshoot and resolve a critical hardware/environmental fault on Cisco PE MAR-PE-13, located at the Marathahalli Core site. The failure domain in question revolves around ASIC memory parity errors, line card crashes, and power supply redundancy loss. The Cisco PE device, operating as a provider edge router, plays a pivotal role in interconnecting multiple networks, ensuring the efficient exchange of data packets. The device's control plane, utilizing the OSPF protocol, maintains a routing table that directs traffic across the network. However, when hardware or environmental issues arise, the device's ability to forward packets is compromised, resulting in packet loss and network instability. The likelihood of such faults occurring is exacerbated by the presence of ASIC memory parity errors, which can lead to line card crashes, and the loss of power supply redundancy, compromising the device's ability to operate in a fault-tolerant manner. This SOP outlines the necessary steps to diagnose and rectify such failures, ensuring minimal network downtime and optimal performance.

Pre-requisite Checks

- Verify the site location and device ID to confirm the issue is specific to MAR-PE-13 at Marathahalli Core.

- Check the redundant link status to ensure alternate paths are available for traffic rerouting.

- Dispatch remote hands to the Marathahalli Core site to facilitate physical access to the device.

- Confirm the device's current software and firmware versions to ensure compatibility and potential for resolution.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Errors	High
Line Card Crashes	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show logging | include ALARM_2961, Expected Output: System logging: ALARM_2961: Critical Hardware/Environmental Fault detected on MAR-PE-13, Description: This command isolates the specific alarm code and logs related to the failure, providing insight into the nature of the issue.
```

```
Command: show inventory | include MAR-PE-13, Expected Output: Cisco PE MAR-PE-13 - Hardware Revision: 1.2, Description: This command verifies the device's hardware revision and model, essential for identifying potential compatibility issues.

Command: show redundancy | include power, Expected Output: Power supply redundancy: 1/N+1, Description: This command checks the power supply redundancy status, verifying the device's ability to operate in a fault-tolerant manner.
```

Resolution Steps

Step 1: Step: Swap out the faulty line card, Command: controller t3 0/0/0 shutdown, Description: This command shuts down the faulty line card, preventing further packet loss and ensuring the device's control plane remains stable.

Step 2: Step: Replace the faulty ASIC memory, Command: controller t3 0/0/0 mem replace, Description: This command initiates the replacement of the faulty ASIC memory, addressing the root cause of the parity errors and crashes.

Step 3: Step: Reboot the device with power supply redundancy, Command: reload in 10, Description: This command reboots the device, ensuring the power supply redundancy is re-established and the device can operate in a fault-tolerant manner.

Post-Check Verification

```
Command: show redundancy | include power, Expected Output: Power supply redundancy: 1/N+1, Description: This command verifies the power supply redundancy status, confirming the device can operate in a fault-tolerant manner.

Command: show inventory | include MAR-PE-13, Expected Output: Cisco PE MAR-PE-13 - Hardware Revision: 1.2, Description: This command verifies the device's hardware revision and model, ensuring compatibility and resolving the initial issue.
```

Roll-back Procedures

- Step: Revert the line card swap, Command: controller t3 0/0/0 no shutdown, Description: This command reverts the line card swap, restoring the original configuration and mitigating the risk of further packet loss.

- Step: Revert the ASIC memory replacement, Command: controller t3 0/0/0 mem revert, Description: This command reverts the ASIC memory replacement, addressing the root cause of the parity errors and crashes.

- Step: Revert the device reboot, Command: no reload, Description: This command reverts the device reboot, ensuring the power supply redundancy is re-established and the device can operate in a fault-tolerant manner.